

### COURSE DESCRIPTION

|               |           |                    |                       |
|---------------|-----------|--------------------|-----------------------|
| Course Number | 0418-221  | Course Title       | Computer Systems      |
| Credit Hours  | 4 (3:3:4) | Course Coordinator | Dr. Mansour Abdulaziz |
| Compulsory    | Yes       | Prerequisite       | 0418-220              |

### Textbook

*Title: Computer Systems: A Programmer's Perspective,*  
*Authors: Randal E. Bryant and David R. O'Hallaron,*  
*Publisher: Pearson, Latest Edition*

### References

*Title: Assembly Language for x86 Processors,*  
*Authors: Kip Irvine,*  
*Publisher: Prentice Hall; 7<sup>th</sup> edition*

### Major Topics Covered in the Course

|   | Topics                                                                                                                                                                                                                 | Hours |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1 | Introduction: C Tour of Computer Systems, language translation stages.                                                                                                                                                 | 3     |
| 2 | Representing and Manipulating Information: information storage, integers, and floating-point representation. Bit-level representation. Bitwise operations, Logical operations.                                         | 9     |
| 3 | Machine-Level Representation of Programs: 64-bit machine programming basics and control, Procedure calls implementations and stack operations and discipline. Structured data alignment (array, structure, and union). | 9     |
| 4 | Processor Architecture: logical gates, Y86-64 ISA, sequential implementation and pipelining.                                                                                                                           | 8     |

|   |                                                                                                                                   |    |
|---|-----------------------------------------------------------------------------------------------------------------------------------|----|
| 5 | Optimizing Program Performance: compiler limitations and improvement techniques: loop, function call, and memory reference usage. | 4  |
| 6 | The Memory Hierarchy: storage technology, locality, memory hierarchy, and cache memories, cache-friendly code.                    | 6  |
| 7 | Exams                                                                                                                             | 3  |
|   | Total                                                                                                                             | 42 |

### Laboratory Work:

|    | Laboratory                                                                         | Weeks |
|----|------------------------------------------------------------------------------------|-------|
| 1  | Assembly Language Tools, Editing, assembling, linking and executing programs       | 1     |
| 2  | Exploring Assembly Data Types<br>Working on Fundamental Instructions               | 2     |
| 3  | Addressing Modes<br>Using Input/ Output Library Function                           | 3     |
| 4  | Implement Logic and arithmetic operations, bit-level representation and operations | 4     |
| 5  | Conditional Processing                                                             | 5     |
| 6  | Procedures and Stack operations                                                    | 6     |
| 7  | Disassembling reverse engineering programs.                                        | 7     |
| 8  | Experience stack discipline and the danger of buffer overflow                      | 8     |
| 9  | Experience with Y86-64 simulator                                                   | 9     |
| 10 | Advanced procedures and macros                                                     | 10    |

|    |                                                                         |    |
|----|-------------------------------------------------------------------------|----|
| 11 | Integer arithmetic and Bit manipulation                                 | 11 |
| 12 | Experience in low-level program optimization                            | 12 |
| 13 | Experience on cache memories benefits in low-level program optimization | 13 |
| 14 | String Primitive Instruction<br>Searching and Sorting Integer Array     | 14 |

### Grading

|        |                    |
|--------|--------------------|
| 15-20% | Midterm 1          |
| 15-20% | Midterm 2          |
| 10-20% | Lab                |
| 0-20%  | Homework / Quizzes |
| 40%    | Final Exam         |

## Course Outcomes

Students should be able to:

1. Understand how to represent and manipulate information.
2. Understand computer programs at the level of instruction set architecture (ISA), and how ISA are designed.
3. Develop assembly programs and understand how high level programs translated to machine language.
4. Design a simple circuit using Boolean Logic and fundamental logic gates building blocks.
5. Optimizing high level programs performance.
6. Understand and appreciate the role of a computer system memory hierarchy including: cache, RAM and secondary storage.

## The relationship between Course Outcomes and Student Outcomes

| Course Outcome | Student Outcomes |   |   |   |   |   |
|----------------|------------------|---|---|---|---|---|
|                | 1                | 2 | 3 | 4 | 5 | 6 |
| 1.             | H                |   |   |   |   |   |
| 2.             | H                | H |   |   |   | M |
| 3.             | H                | M |   |   |   | M |
| 4.             | L                | M |   |   |   |   |
| 5.             | H                | M |   |   |   | L |
| 6.             | H                | H |   |   |   |   |